

9.4 Gram-Schmidt Orthogonalization



Core skills covered in this section:

- ▶ Orthogonalization: converting a set of linearly independent vectors into pairwise orthogonal unit vectors
- ▶ Gram-Schmidt orthogonalization: achieving vector orthogonalization through stepwise projection and normalization.
- ▶ Each step of Gram-Schmidt orthogonalization projects a vector onto the orthogonal complement of the already-obtained orthogonal basis.
- ▶ Every vector can be orthogonally decomposed into "known direction + unique direction".
- ▶ $A = QR$: the finally obtained Q is an orthogonal matrix, and R is an upper triangular coordinate matrix.

In the first three sections of this chapter, we saw various "excellent" properties of orthogonal matrices. This section begins introducing a method for orthogonalizing linearly independent vectors and obtaining an orthogonal matrix. The algorithm we'll use is called Gram-Schmidt orthogonalization, one of several common methods for obtaining orthogonal matrices.

Geometric Perspective

As Figure 1 shows, from a geometric point of view, Gram-Schmidt orthogonalization converts a set of linearly independent vectors (matrix A 's column vectors) into pairwise orthogonal unit vectors (matrix Q 's column vectors).

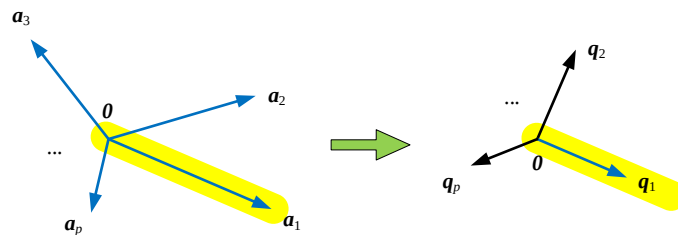


Figure 1. The Geometric Meaning Behind Gram-Schmidt Orthogonalization

⚠ Note that Figure 1 specifically highlights in yellow that column vector a_1 is parallel to q_1 , which is one of the important characteristics of Gram-Schmidt orthogonalization.

As Figure 2 shows, from a matrix transformation point of view, Gram-Schmidt orthogonalization converts a full column rank matrix A into a semi-orthogonal matrix Q .

From a basis point of view, Gram-Schmidt orthogonalization converts the non-orthogonal basis $[\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_p]$ into the orthonormal basis $[\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_p]$.

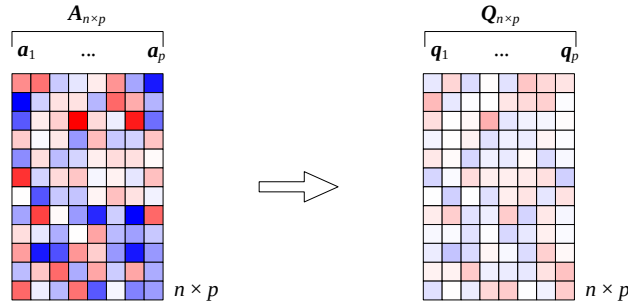


Figure 2. Full Column Rank Matrix $\mathbf{A}$ Converted Into Semi-Orthogonal Matrix $\mathbf{Q}$

The Orthogonalization Process

Using vector projection, the Gram-Schmidt orthogonalization process is

$$\begin{aligned}\eta_1 &= \mathbf{a}_1 \\ \eta_2 &= \mathbf{a}_2 - \text{proj}_{\eta_1}(\mathbf{a}_2) \\ \eta_3 &= \mathbf{a}_3 - \text{proj}_{\eta_1}(\mathbf{a}_3) - \text{proj}_{\eta_2}(\mathbf{a}_3) \\ &\vdots \\ \eta_p &= \mathbf{a}_p - \sum_{j=1}^{p-1} \text{proj}_{\eta_j}(\mathbf{a}_p)\end{aligned}$$

Finally, normalizing gives the orthonormal vectors

$$\mathbf{q}_1 = \frac{\eta_1}{\|\eta_1\|}, \quad \mathbf{q}_2 = \frac{\eta_2}{\|\eta_2\|}, \quad \mathbf{q}_3 = \frac{\eta_3}{\|\eta_3\|}, \quad \dots, \quad \mathbf{q}_p = \frac{\eta_p}{\|\eta_p\|}$$

In the result, $\mathbf{q}_i$ is a unit vector, and $\mathbf{q}_i$ and $\mathbf{q}_j$ are pairwise orthogonal.

It's worth emphasizing again that what's special about the orthonormal basis $[\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3, \dots, \mathbf{q}_p]$ is that $\mathbf{q}_1$ is parallel to $\mathbf{a}_1$. This book will introduce other algorithms for obtaining an orthonormal basis later; please compare them carefully.

⚠ Note that you absolutely should not just memorize this Gram-Schmidt orthogonalization process by rote. We'll immediately use an example to show you the geometric perspective of this orthogonalization process, which will greatly help you understand the process above.

Orthogonal Projection

Based on the orthogonal projection introduced earlier in this book, the orthogonal projection used in the second equation above can be written as

$$\text{proj}_{\eta_1}(\mathbf{a}_2) = \frac{\eta_1 @ \eta_1^T}{\eta_1^T @ \eta_1} @ \mathbf{a}_2 = \mathbf{q}_1 @ \mathbf{q}_1^T @ \mathbf{a}_2$$

After obtaining η_1 , first normalize it to get $\mathbf{q}_1$; this way, when computing η_2 , we can use

$$\eta_2 = \mathbf{a}_2 - \mathbf{q}_1 @ \mathbf{q}_1^T @ \mathbf{a}_2 = (\mathbf{I} - \mathbf{q}_1 @ \mathbf{q}_1^T) @ \mathbf{a}_2$$

The geometric meaning of the equation above is that vector $\mathbf{a}_2$ is projected onto the orthogonal complement of $\text{span}(\mathbf{q}_1)$; the resulting projection η_2 must be perpendicular to $\mathbf{q}_1$.

After obtaining η_2 , immediately normalize it to get $\mathbf{q}_2$; computing η_3 also uses orthogonal projection

$$\eta_3 = \mathbf{a}_3 - \mathbf{q}_1 @ \mathbf{q}_1^T @ \mathbf{a}_3 - \mathbf{q}_2 @ \mathbf{q}_2^T @ \mathbf{a}_3 = (\mathbf{I} - \mathbf{q}_1 @ \mathbf{q}_1^T - \mathbf{q}_2 @ \mathbf{q}_2^T) @ \mathbf{a}_3$$

From a geometric point of view, the equation above is equivalent to projecting vector $\mathbf{a}_3$ onto the orthogonal complement of $\text{span}(\mathbf{q}_1, \mathbf{q}_2)$; the resulting projection η_3 is perpendicular to both $\mathbf{q}_1$ and $\mathbf{q}_2$.

By analogy, using the same method we can orthogonalize (full column rank, i.e., linearly independent) $[\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3, \dots, \mathbf{a}_p]$ to obtain $[\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3, \dots, \mathbf{q}_p]$.

A Worked Example

Below, let's use Gram-Schmidt orthogonalization step by step on the following square matrix's column vectors

$$\mathbf{A} = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 0 & 2 \\ 0 & 2 & 2 \end{bmatrix}$$

$\mathbf{A}$'s column vectors

$$\mathbf{a}_1 = \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix}, \quad \mathbf{a}_2 = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}, \quad \mathbf{a}_3 = \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix}$$

As Figure 3 shows, $\mathbf{a}_1$, $\mathbf{a}_2$, and $\mathbf{a}_3$ are not unit vectors, and are not pairwise orthogonal; however, $\mathbf{a}_1$, $\mathbf{a}_2$, and $\mathbf{a}_3$ are linearly independent, so they can form a non-orthogonal basis, and likewise span three-dimensional space.

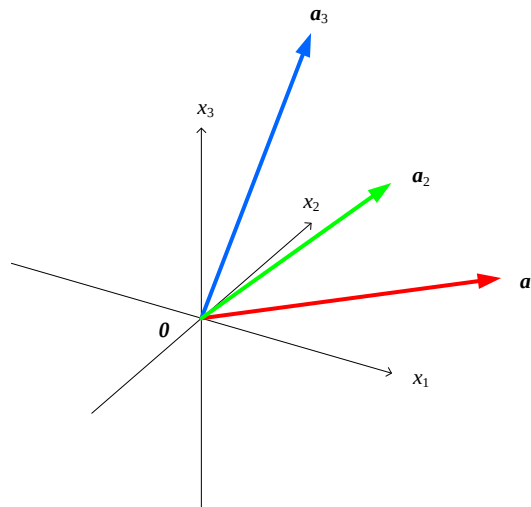


Figure 3. $\mathbf{a}_1$, $\mathbf{a}_2$, and $\mathbf{a}_3$ Specific Positions

Step 1: $\mathbf{a}_1 \rightarrow \mathbf{q}_1$

Step 1 is the simplest; to obtain $\boldsymbol{\eta}_1$, we only need $\boldsymbol{\eta}_1 = \mathbf{a}_1$.

Then, normalizing the nonzero vector $\boldsymbol{\eta}_1$ gives $\mathbf{q}_1$

$$\mathbf{q}_1 = \frac{\boldsymbol{\eta}_1}{\|\boldsymbol{\eta}_1\|} = \frac{\mathbf{a}_1}{\|\mathbf{a}_1\|} = \begin{bmatrix} \sqrt{2}/2 \\ \sqrt{2}/2 \\ 0 \end{bmatrix}$$

Specifically, as Figure 4 shows; clearly, $\mathbf{q}_1$ and $\mathbf{a}_1$ ($\boldsymbol{\eta}_1$) point in the same direction.

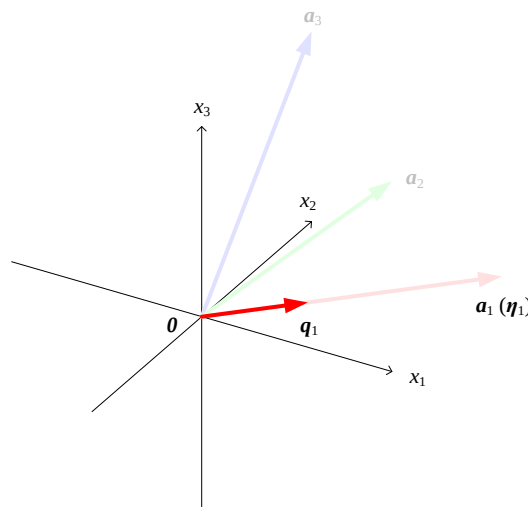
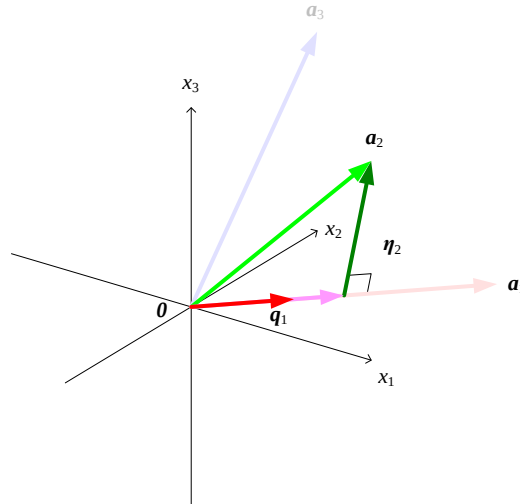


Figure 4. Normalizing $\mathbf{a}_1$ ($\boldsymbol{\eta}_1$)

Step 2: $\mathbf{a}_2 \rightarrow \mathbf{q}_2$

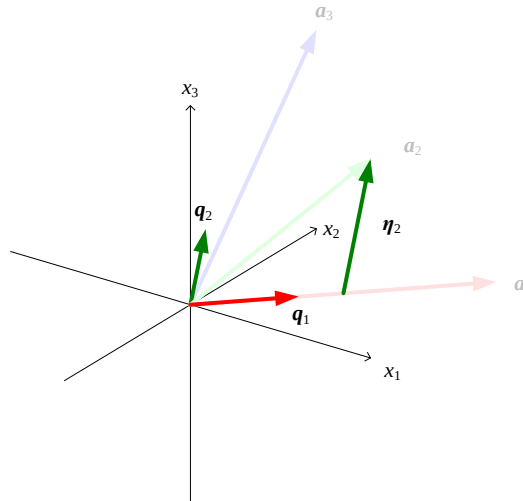
After obtaining $\mathbf{q}_1$, use orthogonal projection to compute $\boldsymbol{\eta}_2$

$$\begin{aligned} \boldsymbol{\eta}_2 &= (\mathbf{I} - \mathbf{q}_1 @ \mathbf{q}_1^T) @ \mathbf{a}_2 \\ &= \left(\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} \sqrt{2}/2 \\ \sqrt{2}/2 \\ 0 \end{bmatrix} @ \begin{bmatrix} \sqrt{2}/2 & \sqrt{2}/2 & 0 \end{bmatrix}^T \right) @ \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix} \\ &= \begin{bmatrix} 1/2 & -1/2 & 0 \\ -1/2 & 1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} @ \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} \end{aligned}$$

Figure 5. Orthogonal Projection Computation of η_2

Then, as Figure 6 shows, normalizing η_2 gives q_2

$$q_2 = \frac{\eta_2}{\|\eta_2\|} = \frac{\sqrt{6}}{6} \times \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} = \begin{bmatrix} \sqrt{6}/6 \\ -\sqrt{6}/6 \\ \sqrt{6}/3 \end{bmatrix}$$

Figure 6. Normalizing η_2

We're done with the computation; now let's do some "extra" analysis.

As Figure 7 shows, this step is equivalent to orthogonally decomposing a_2 into two parts $q_1 @ q_1^T @ a_2$ and η_2 , corresponding to

$$a_2 = q_1 @ q_1^T @ a_2 + \eta_2$$

That is to say, a_2 is orthogonally decomposed into two components: one parallel to q_1 (a_1, η_1), and one perpendicular to q_1 .

And after removing from $\mathbf{a}_2$ the component in the same direction as $\mathbf{a}_1$, the remaining component perpendicular to $\mathbf{a}_1$ ($\boldsymbol{\eta}_2$) is precisely the "unique" direction contributed by $\mathbf{a}_2$!

With $\mathbf{q}_2$ in hand, $\boldsymbol{\eta}_2$ can be written as

$$\boldsymbol{\eta}_2 = \mathbf{q}_2 @ \mathbf{q}_2^T @ \mathbf{a}_2$$

As Figure 7 shows, the orthogonal decomposition of vector $\mathbf{a}_2$ can be written as

$$\begin{aligned}\mathbf{a}_2 &= \mathbf{q}_1 @ \mathbf{q}_1^T @ \mathbf{a}_2 + \mathbf{q}_2 @ \mathbf{q}_2^T @ \mathbf{a}_2 \\ &= (\mathbf{q}_1 @ \mathbf{q}_1^T + \mathbf{q}_2 @ \mathbf{q}_2^T) @ \mathbf{a}_2\end{aligned}$$

⚠ Note that $\mathbf{q}_1 @ \mathbf{q}_1^T + \mathbf{q}_2 @ \mathbf{q}_2^T$ is not full rank, so the square matrix is not invertible!

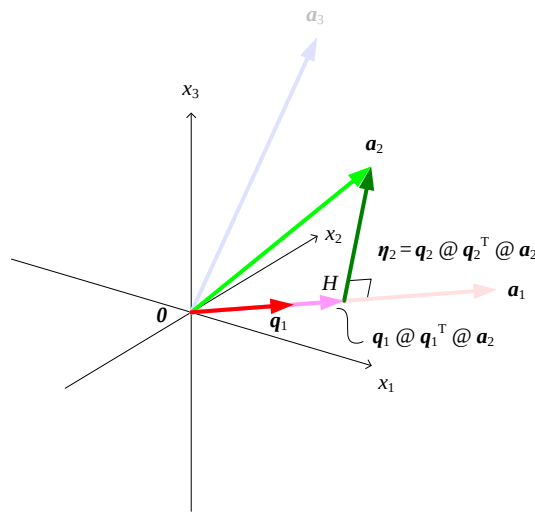


Figure 7. Orthogonal Decomposition of $\mathbf{a}_2$

Computing $\mathbf{q}_1 @ \mathbf{q}_1^T @ \mathbf{a}_2$,

$$\begin{aligned}\mathbf{q}_1 @ \mathbf{q}_1^T @ \mathbf{a}_2 &= \begin{bmatrix} \sqrt{2}/2 \\ \sqrt{2}/2 \\ 0 \end{bmatrix} @ \begin{bmatrix} \sqrt{2}/2 \\ \sqrt{2}/2 \\ 0 \end{bmatrix}^T @ \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix} \\ &= \begin{bmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 1/2 & 0 \\ 0 & 0 & 0 \end{bmatrix} @ \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}\end{aligned}$$

In the expression $\mathbf{q}_1 @ \mathbf{q}_1^T @ \mathbf{a}_2$, $\mathbf{q}_1^T @ \mathbf{a}_2$ is the scalar projection, i.e., the coordinate of vector $\mathbf{a}_2$ along the direction vector (unit vector) $\mathbf{q}_1$; multiplying again by $\mathbf{q}_1$ then gives the vector projection. This is the "double projection" mentioned earlier!

$\mathbf{q}_1 @ \mathbf{q}_1^T @ \mathbf{a}_2$ is the coordinate value of point H in Figure 7!

Clearly, $\mathbf{q}_1 @ \mathbf{q}_1^T @ \mathbf{a}_2$ is parallel to $\mathbf{a}_1$

$$\mathbf{q}_1 @ \mathbf{q}_1^T @ \mathbf{a}_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \frac{\mathbf{a}_1}{2} = \frac{1}{2} \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix}$$

Verify that $\boldsymbol{\eta}_2$ and $\mathbf{q}_1 @ \mathbf{q}_1^T @ \mathbf{a}_2$ are orthogonal

$$\boldsymbol{\eta}_2 \cdot (\mathbf{q}_1 @ \mathbf{q}_1^T @ \mathbf{a}_2) = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = 0$$

Now verify that $\boldsymbol{\eta}_2$ and $\mathbf{q}_1 @ \mathbf{q}_1^T @ \mathbf{a}_2$ combine to give $\mathbf{a}_2$, i.e.,

$$\boldsymbol{\eta}_2 + \mathbf{q}_1 @ \mathbf{q}_1^T @ \mathbf{a}_2 = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix} = \mathbf{a}_2$$

Step 3: $\mathbf{a}_3 \rightarrow \mathbf{q}_3$

After obtaining $\mathbf{q}_1$ and $\mathbf{q}_2$, use orthogonal projection to compute $\boldsymbol{\eta}_3$

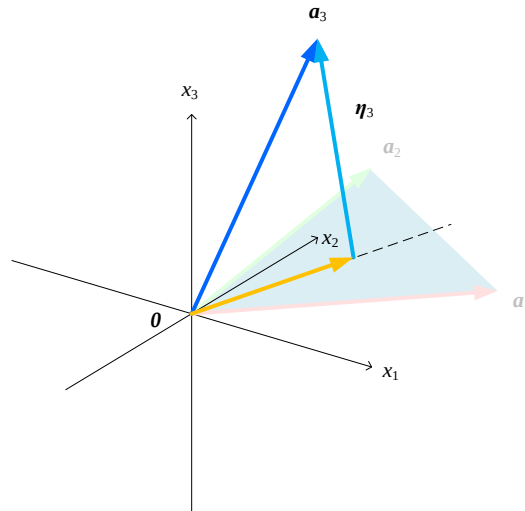
$$\begin{aligned} \boldsymbol{\eta}_3 &= (\mathbf{I} - \mathbf{q}_1 @ \mathbf{q}_1^T - \mathbf{q}_2 @ \mathbf{q}_2^T) @ \mathbf{a}_3 \\ &= \left(\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} \sqrt{2}/2 \\ \sqrt{2}/2 \\ 0 \end{bmatrix} @ \begin{bmatrix} \sqrt{2}/2 \\ \sqrt{2}/2 \\ 0 \end{bmatrix}^T - \begin{bmatrix} \sqrt{6}/6 \\ -\sqrt{6}/6 \\ \sqrt{6}/3 \end{bmatrix} @ \begin{bmatrix} \sqrt{6}/6 \\ -\sqrt{6}/6 \\ \sqrt{6}/3 \end{bmatrix}^T \right) @ \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix} \\ &= \left(\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 1/2 & 0 \\ 0 & 0 & 0 \end{bmatrix} - \begin{bmatrix} 1/6 & -1/6 & 1/3 \\ -1/6 & 1/6 & -1/3 \\ 1/3 & -1/3 & 2/3 \end{bmatrix} \right) @ \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix} \\ &= \begin{bmatrix} 1/3 & -1/3 & -1/3 \\ -1/3 & 1/3 & 1/3 \\ -1/3 & 1/3 & 1/3 \end{bmatrix} @ \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} -4/3 \\ 4/3 \\ 4/3 \end{bmatrix} \end{aligned}$$

From a geometric point of view, step 3 projects $\mathbf{a}_3$ onto the plane spanned by $[\boldsymbol{\eta}_1, \boldsymbol{\eta}_2]$. And $\boldsymbol{\eta}_3$ is $\mathbf{a}_3$'s component that is not on the $[\boldsymbol{\eta}_1, \boldsymbol{\eta}_2]$ plane.

⚠ Note that $\text{span}(\boldsymbol{\eta}_1, \boldsymbol{\eta}_2)$ is equivalent to $\text{span}(\mathbf{q}_1, \mathbf{q}_2)$.

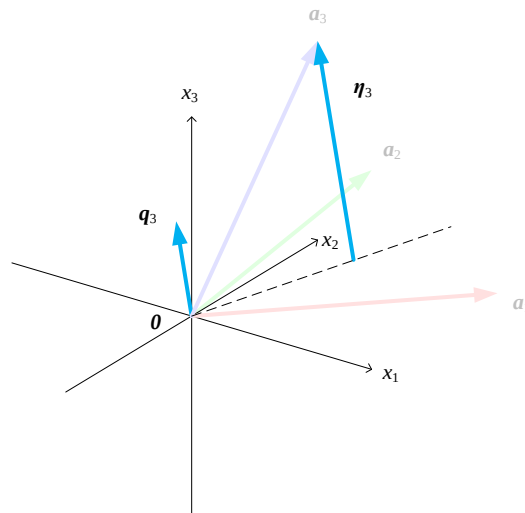
Clearly, $\boldsymbol{\eta}_3$ is perpendicular to $\text{span}(\boldsymbol{\eta}_1, \boldsymbol{\eta}_2)$, which is to say $\boldsymbol{\eta}_3$ is perpendicular to both $\boldsymbol{\eta}_1$ and $\boldsymbol{\eta}_2$. This is to say, $\text{span}(\boldsymbol{\eta}_3)$ and $\text{span}(\boldsymbol{\eta}_1, \boldsymbol{\eta}_2)$ are orthogonal complements of each other.

As Figure 8 shows, after removing from $\mathbf{a}_3$ the components parallel to $\mathbf{a}_1$ and $\mathbf{a}_2$, the remaining $\boldsymbol{\eta}_3$ is $\mathbf{a}_3$'s "unique" contribution.

Figure 8. Orthogonal Projection Computation of η_3

Then, as Figure 9 shows, normalizing η_3 gives q_3

$$q_3 = \frac{\eta_3}{\|\eta_3\|} = \begin{bmatrix} -\sqrt{3}/3 \\ \sqrt{3}/3 \\ \sqrt{3}/3 \end{bmatrix}$$

Figure 9. Normalizing η_3 

LA_09_04_01.ipynb implements the above computations; please analyze the code line by line against this section's content.

At this point, as Figure 10 shows, we've completed the orthogonalization of $[a_1, a_2, a_3]$, obtaining $[q_1, q_2, q_3]$.

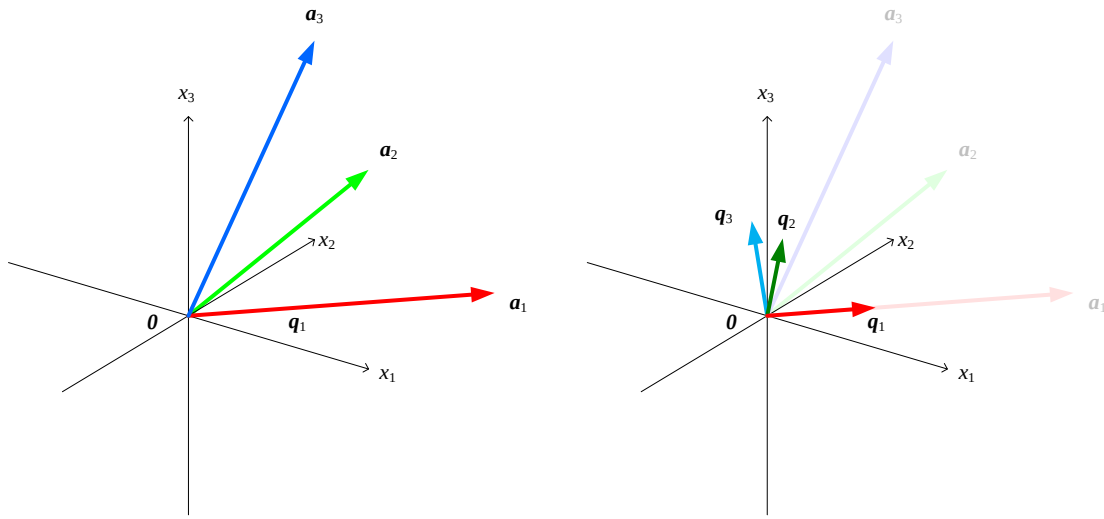


Figure 10. Orthogonalizing $[a_1, a_2, a_3]$ to Obtain $[q_1, q_2, q_3]$

However, our analysis isn't finished yet!

As Figure 11 shows, this step is equivalent to orthogonally decomposing a_3 into two parts $(q_1 @ q_1^T + q_2 @ q_2^T) @ a_3$ and η_3 , corresponding to

$$a_3 = (q_1 @ q_1^T + q_2 @ q_2^T) @ a_3 + \eta_3$$

That is to say, a_3 is orthogonally decomposed into two components: one on the $\text{span}(q_1, q_2)$ plane, and one perpendicular to $\text{span}(q_1, q_2)$.

And after removing from a_3 the component on the $\text{span}(q_1, q_2)$ plane, the remaining component perpendicular to $\text{span}(q_1, q_2)$ (η_3) is precisely the "unique" direction contributed by a_3 !

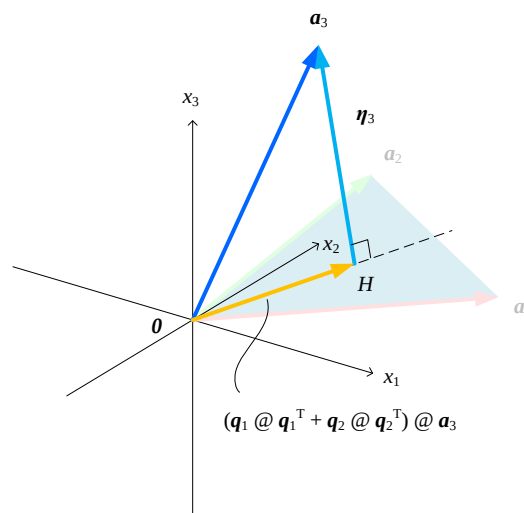


Figure 11. Orthogonal Decomposition of a_3

As Figure 12 shows, $(\mathbf{q}_1 @ \mathbf{q}_1^T + \mathbf{q}_2 @ \mathbf{q}_2^T) @ \mathbf{a}_3$ can be further orthogonally decomposed into $(\mathbf{q}_1 @ \mathbf{q}_1^T) @ \mathbf{a}_3$ and $(\mathbf{q}_2 @ \mathbf{q}_2^T) @ \mathbf{a}_3$.

Computing $\mathbf{q}_1 @ \mathbf{q}_1^T @ \mathbf{a}_3$,

$$\begin{aligned}\mathbf{q}_1 @ \mathbf{q}_1^T @ \mathbf{a}_3 &= \begin{bmatrix} \sqrt{2}/2 \\ \sqrt{2}/2 \\ 0 \end{bmatrix} @ \begin{bmatrix} \sqrt{2}/2 \\ \sqrt{2}/2 \\ 0 \end{bmatrix}^T @ \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix} \\ &= \begin{bmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 1/2 & 0 \\ 0 & 0 & 0 \end{bmatrix} @ \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}\end{aligned}$$

As Figure 12 shows, from a geometric point of view, the expression $\mathbf{q}_1 @ \mathbf{q}_1^T @ \mathbf{a}_3$ represents the orthogonal projection of $\mathbf{a}_3$ onto the direction vector $\mathbf{q}_1$.

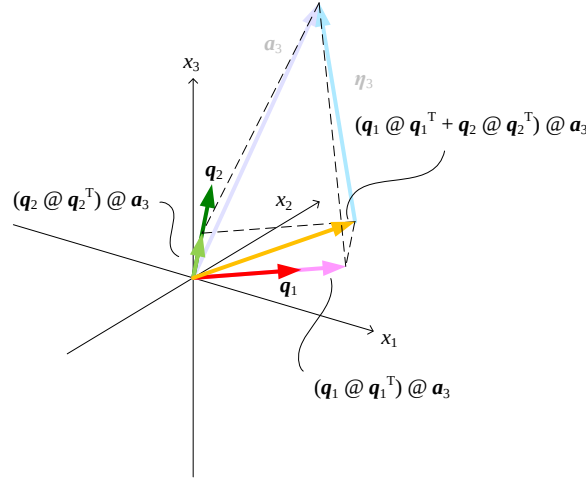


Figure 12. Orthogonal Decomposition of $(\mathbf{q}_1 @ \mathbf{q}_1^T + \mathbf{q}_2 @ \mathbf{q}_2^T) @ \mathbf{a}_3$

Computing $\mathbf{q}_2 @ \mathbf{q}_2^T @ \mathbf{a}_3$,

$$\begin{aligned}\mathbf{q}_2 @ \mathbf{q}_2^T @ \mathbf{a}_3 &= \begin{bmatrix} \sqrt{6}/6 \\ -\sqrt{6}/6 \\ \sqrt{6}/3 \end{bmatrix} @ \begin{bmatrix} \sqrt{6}/6 \\ -\sqrt{6}/6 \\ \sqrt{6}/3 \end{bmatrix}^T @ \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix} \\ &= \begin{bmatrix} 1/6 & -1/6 & 1/3 \\ -1/6 & 1/6 & -1/3 \\ 1/3 & -1/3 & 2/3 \end{bmatrix} @ \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 1/3 \\ -1/3 \\ 2/3 \end{bmatrix}\end{aligned}$$

As Figure 12 shows, from a geometric point of view, the expression $\mathbf{q}_2 @ \mathbf{q}_2^T @ \mathbf{a}_3$ represents the orthogonal projection of $\mathbf{a}_3$ onto the direction vector $\mathbf{q}_2$.

As Figure 13 shows, $\boldsymbol{\eta}_3$, $\mathbf{q}_1 @ \mathbf{q}_1^T @ \mathbf{a}_3$, and $\mathbf{q}_2 @ \mathbf{q}_2^T @ \mathbf{a}_3$ combine to give $\mathbf{a}_3$, i.e.,

$$\eta_3 + \mathbf{q}_1 @ \mathbf{q}_1^T @ \mathbf{a}_3 + \mathbf{q}_2 @ \mathbf{q}_2^T @ \mathbf{a}_3 = \begin{bmatrix} -4/3 \\ 4/3 \\ 4/3 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 1/3 \\ -1/3 \\ 2/3 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix} = \mathbf{a}_3$$

Please verify on your own that η_3 , $\mathbf{q}_1 @ \mathbf{q}_1^T @ \mathbf{a}_3$, and $\mathbf{q}_2 @ \mathbf{q}_2^T @ \mathbf{a}_3$ are pairwise orthogonal.

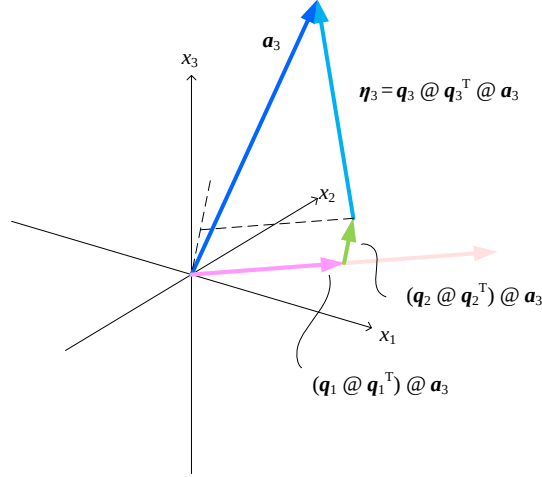


Figure 13. Three Vectors Connected End-to-End Restore $\mathbf{a}_3$

With $\mathbf{q}_3$ in hand, η_3 can actually be written as

$$\eta_3 = \mathbf{q}_3 @ \mathbf{q}_3^T @ \mathbf{a}_3$$

In this way, we obtain

$$\mathbf{a}_3 = (\mathbf{q}_1 @ \mathbf{q}_1^T + \mathbf{q}_2 @ \mathbf{q}_2^T + \mathbf{q}_3 @ \mathbf{q}_3^T) @ \mathbf{a}_3$$

This contains the familiar

$$\mathbf{q}_1 @ \mathbf{q}_1^T + \mathbf{q}_2 @ \mathbf{q}_2^T + \mathbf{q}_3 @ \mathbf{q}_3^T = \mathbf{I}$$

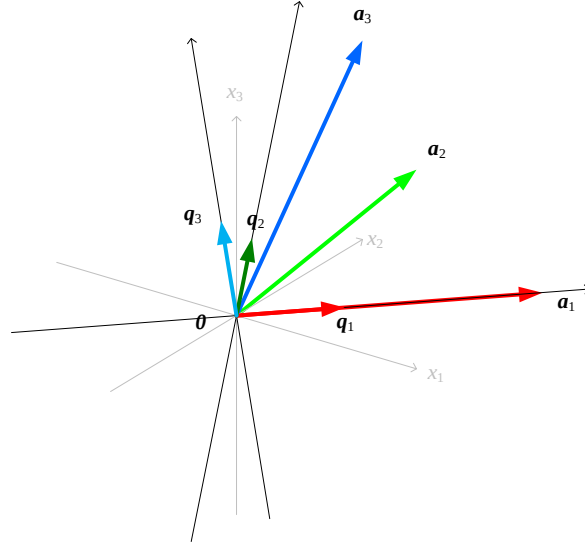
$[\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3]$ Coordinates

In the $[\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3]$ coordinate system, we know the coordinates of $\mathbf{a}_1$, $\mathbf{a}_2$, and $\mathbf{a}_3$, i.e.,

$$\mathbf{a}_1 = \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix}, \quad \mathbf{a}_2 = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}, \quad \mathbf{a}_3 = \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix}$$

And $[\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3]$ is an orthonormal basis; the parallelepiped it spans is a rotated unit cube.

As Figure 14 shows, $\mathbf{a}_1$, $\mathbf{a}_2$, and $\mathbf{a}_3$ of course also have their own coordinates in $[\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3]$!

Figure 14. $\mathbf{a}_1$, $\mathbf{a}_2$, and $\mathbf{a}_3$ in the Orthonormal Basis $[\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3]$

Below, let's compute these coordinate values one by one.

$\mathbf{a}_1$ in $[\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3]$ Coordinates

Let $\mathbf{a}_1$'s coordinates in $[\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3]$ be $(r_{1,1}, r_{2,1}, r_{3,1})$, that is to say $\mathbf{a}_1$ can be written as the following linear combination

$$\mathbf{a}_1 = r_{1,1}\mathbf{q}_1 + r_{2,1}\mathbf{q}_2 + r_{3,1}\mathbf{q}_3$$

Written in matrix multiplication form

$$\mathbf{a}_1 = [\mathbf{q}_1 \quad \mathbf{q}_2 \quad \mathbf{q}_3] \begin{bmatrix} r_{1,1} \\ r_{2,1} \\ r_{3,1} \end{bmatrix}$$

Based on scalar projection, vector $\mathbf{a}_1$'s coordinate along the direction vector (unit vector) $\mathbf{q}_1$ is the inner product of the two, i.e.,

$$r_{1,1} = \mathbf{a}_1 \cdot \mathbf{q}_1 = \mathbf{q}_1 \cdot \mathbf{a}_1 = \langle \mathbf{a}_1, \mathbf{q}_1 \rangle = \mathbf{a}_1^T @ \mathbf{q}_1 = \mathbf{q}_1^T @ \mathbf{a}_1$$

Substituting in the specific values

$$r_{1,1} = \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} \sqrt{2}/2 \\ \sqrt{2}/2 \\ 0 \end{bmatrix} = 2\sqrt{2}$$

This is $\mathbf{a}_1$'s coordinate on the $\mathbf{q}_1$ axis (scalar projection).

Similarly, we can obtain

$$\begin{aligned} r_{2,1} &= \mathbf{a}_1 \cdot \mathbf{q}_2 = \mathbf{q}_2 \cdot \mathbf{a}_1 = \langle \mathbf{a}_1, \mathbf{q}_2 \rangle = \mathbf{a}_1^T @ \mathbf{q}_2 = \mathbf{q}_2^T @ \mathbf{a}_1 \\ r_{3,1} &= \mathbf{a}_1 \cdot \mathbf{q}_3 = \mathbf{q}_3 \cdot \mathbf{a}_1 = \langle \mathbf{a}_1, \mathbf{q}_3 \rangle = \mathbf{a}_1^T @ \mathbf{q}_3 = \mathbf{q}_3^T @ \mathbf{a}_1 \end{aligned}$$

Since $\mathbf{a}_1$ is parallel to $\mathbf{q}_1$, and $\mathbf{q}_1$, $\mathbf{q}_2$, and $\mathbf{q}_3$ are pairwise orthogonal; clearly, $\mathbf{a}_1$'s scalar projection onto $\mathbf{q}_2$ and $\mathbf{q}_3$ is 0.

This means $r_{2,1} = 0$, $r_{3,1} = 0$, and this can be written as

$$\mathbf{a}_1 = [\mathbf{q}_1 \quad \mathbf{q}_2 \quad \mathbf{q}_3] \begin{bmatrix} r_{1,1} \\ 0 \\ 0 \end{bmatrix}$$

Please use the vector inner product, substituting specific values to compute $r_{2,1}$ and $r_{3,1}$, and check whether both are 0.

Figure 15 shows $\mathbf{a}_1$'s coordinates in $[\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3]$.

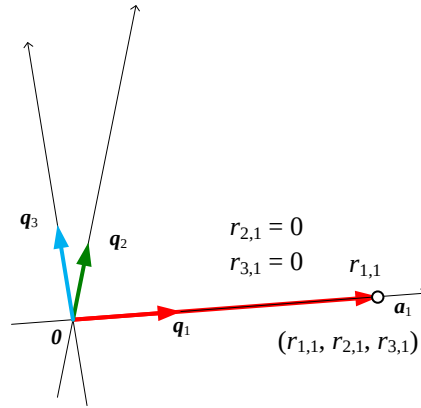


Figure 15. $\mathbf{a}_1$'s Coordinates in $[\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3]$

$\mathbf{a}_2$ in $[\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3]$ Coordinates

Let $\mathbf{a}_2$'s coordinates in $[\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3]$ be $(r_{1,2}, r_{2,2}, r_{3,2})$, that is to say $\mathbf{a}_2$ can be written as the following linear combination

$$\mathbf{a}_2 = r_{1,2}\mathbf{q}_1 + r_{2,2}\mathbf{q}_2 + r_{3,2}\mathbf{q}_3$$

Written in matrix multiplication form

$$\mathbf{a}_2 = [\mathbf{q}_1 \quad \mathbf{q}_2 \quad \mathbf{q}_3] \begin{bmatrix} r_{1,2} \\ r_{2,2} \\ r_{3,2} \end{bmatrix}$$

Likewise, based on scalar projection, vector $\mathbf{a}_2$'s coordinate along the direction vector $\mathbf{q}_1$ is the inner product of the two, i.e.,

$$r_{1,2} = \mathbf{a}_2 \cdot \mathbf{q}_1 = \mathbf{q}_1 \cdot \mathbf{a}_2 = \langle \mathbf{a}_2, \mathbf{q}_1 \rangle = \mathbf{a}_2^T @ \mathbf{q}_1 = \mathbf{q}_1^T @ \mathbf{a}_2$$

Substituting in the specific values

$$r_{1,2} = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} \sqrt{2}/2 \\ \sqrt{2}/2 \\ 0 \end{bmatrix} = \sqrt{2}$$

This is $\mathbf{a}_2$'s coordinate on the $\mathbf{q}_1$ axis (scalar projection).

Similarly, we can obtain

$$r_{2,2} = \mathbf{a}_2 \cdot \mathbf{q}_2 = \mathbf{q}_2 \cdot \mathbf{a}_2 = \langle \mathbf{a}_2, \mathbf{q}_2 \rangle = \mathbf{a}_2^T @ \mathbf{q}_2 = \mathbf{q}_2^T @ \mathbf{a}_2$$

Substituting in the specific values

$$r_{2,2} = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} \sqrt{6}/6 \\ -\sqrt{6}/6 \\ \sqrt{6}/3 \end{bmatrix} = \sqrt{6}$$

This is $\mathbf{a}_2$'s coordinate on the $\mathbf{q}_2$ axis (scalar projection).

Since $\mathbf{a}_2$ is in $\text{span}(\mathbf{q}_1, \mathbf{q}_2)$, $\mathbf{a}_2$'s scalar projection onto $\mathbf{q}_3$ is 0, which means $r_{3,2} = 0$. This can be written as

$$\mathbf{a}_2 = [\mathbf{q}_1 \quad \mathbf{q}_2 \quad \mathbf{q}_3] \begin{bmatrix} r_{1,2} \\ r_{2,2} \\ 0 \end{bmatrix}$$

Please use the vector inner product, substituting specific values to compute $r_{3,2}$, and check whether it is 0.

Figure 16 shows $\mathbf{a}_2$'s coordinates in $[\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3]$.

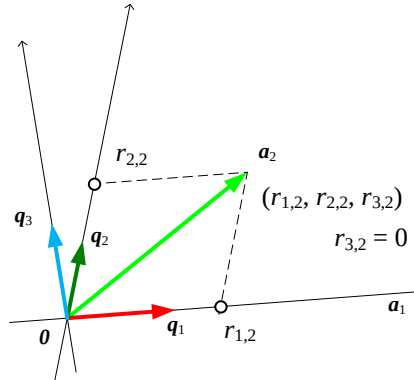


Figure 16. $\mathbf{a}_2$'s Coordinates in $[\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3]$

$\mathbf{a}_3$ in $[\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3]$ Coordinates

Let $\mathbf{a}_3$'s coordinates in $[\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3]$ be $(r_{1,3}, r_{2,3}, r_{3,3})$, that is to say $\mathbf{a}_3$ can be written as the following linear combination

$$\mathbf{a}_3 = r_{1,3}\mathbf{q}_1 + r_{2,3}\mathbf{q}_2 + r_{3,3}\mathbf{q}_3$$

Based on scalar projection, vector $\mathbf{a}_3$'s coordinate along the direction vector $\mathbf{q}_1$ is the inner product of the two, i.e.,

$$r_{1,3} = \mathbf{a}_3 \cdot \mathbf{q}_1 = \mathbf{q}_1 \cdot \mathbf{a}_3 = \langle \mathbf{a}_3, \mathbf{q}_1 \rangle = \mathbf{a}_3^T @ \mathbf{q}_1 = \mathbf{q}_1^T @ \mathbf{a}_3$$

Substituting in the specific values

$$r_{1,3} = \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} \sqrt{2}/2 \\ \sqrt{2}/2 \\ 0 \end{bmatrix} = \sqrt{2}$$

This is $\mathbf{a}_3$'s coordinate on the $\mathbf{q}_1$ axis (scalar projection).

Similarly, we can obtain

$$r_{2,3} = \mathbf{a}_3 \cdot \mathbf{q}_2 = \mathbf{q}_2 \cdot \mathbf{a}_3 = \langle \mathbf{a}_3, \mathbf{q}_2 \rangle = \mathbf{a}_3^T @ \mathbf{q}_2 = \mathbf{q}_2^T @ \mathbf{a}_3$$

Substituting in the specific values

$$r_{2,3} = \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} \sqrt{6}/6 \\ -\sqrt{6}/6 \\ \sqrt{6}/3 \end{bmatrix} = \sqrt{6}/3$$

This is $\mathbf{a}_3$'s coordinate on the $\mathbf{q}_2$ axis (scalar projection).

Similarly, we can obtain

$$r_{3,3} = \mathbf{a}_3 \cdot \mathbf{q}_3 = \mathbf{q}_3 \cdot \mathbf{a}_3 = \langle \mathbf{a}_3, \mathbf{q}_3 \rangle = \mathbf{a}_3^T @ \mathbf{q}_3 = \mathbf{q}_3^T @ \mathbf{a}_3$$

Substituting in the specific values

$$r_{3,3} = \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} -\sqrt{3}/3 \\ \sqrt{3}/3 \\ \sqrt{3}/3 \end{bmatrix} = 4\sqrt{3}/3$$

Written in matrix multiplication form

$$\mathbf{a}_3 = [\mathbf{q}_1 \quad \mathbf{q}_2 \quad \mathbf{q}_3] \begin{bmatrix} r_{1,3} \\ r_{2,3} \\ r_{3,3} \end{bmatrix}$$

Figure 17 shows $\mathbf{a}_3$'s coordinates in $[\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3]$.

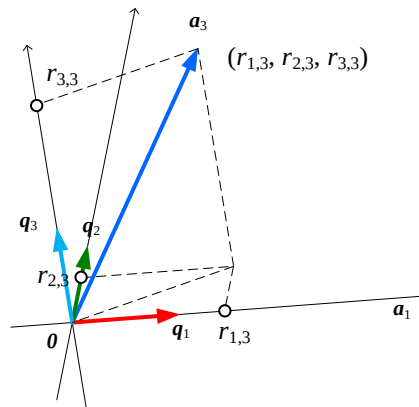


Figure 17. $\mathbf{a}_3$'s Coordinates in $[\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3]$

A Set of Orthogonal Decompositions

In this way, we obtain a set of vector decompositions:

$$\begin{aligned} \mathbf{a}_1 &= \mathbf{q}_1 @ \mathbf{q}_1^T @ \mathbf{a}_1 \\ \mathbf{a}_2 &= \mathbf{q}_1 @ \mathbf{q}_1^T @ \mathbf{a}_2 + \mathbf{q}_2 @ \mathbf{q}_2^T @ \mathbf{a}_2 \\ \mathbf{a}_3 &= \mathbf{q}_1 @ \mathbf{q}_1^T @ \mathbf{a}_3 + \mathbf{q}_2 @ \mathbf{q}_2^T @ \mathbf{a}_3 + \mathbf{q}_3 @ \mathbf{q}_3^T @ \mathbf{a}_3 \end{aligned}$$

In the equations above, we once again see orthogonal projection and vector orthogonal decomposition!

There's also an interesting detail here

$$\mathbf{a}_1 = (\mathbf{q}_1 @ \mathbf{q}_1^T) @ \mathbf{a}_1$$

This actually leads into eigenvalue decomposition $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$.

Simply put, for a square matrix $\mathbf{A}$, eigenvalue decomposition finds a set of nonzero vectors $\mathbf{v}$ and corresponding scalars λ such that $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$, i.e., the square matrix $\mathbf{A}$ only stretches these directions without changing their direction (same or opposite direction).

⚠ Eigenvalue decomposition is one of the topics this book will focus on and explain in detail later.

The Third Perspective of Matrix Multiplication: Column Vector Combination

Arranging the three matrix multiplication equations above in sequence gives

$$\underbrace{\begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \mathbf{a}_3 \end{bmatrix}}_{\mathbf{A}} = \underbrace{\begin{bmatrix} \mathbf{q}_1 & \mathbf{q}_2 & \mathbf{q}_3 \end{bmatrix}}_{\mathbf{Q}} \underbrace{\begin{bmatrix} \begin{bmatrix} r_{1,1} \\ 0 \\ 0 \end{bmatrix} & \begin{bmatrix} r_{1,2} \\ r_{2,2} \\ 0 \end{bmatrix} & \begin{bmatrix} r_{1,3} \\ r_{2,3} \\ r_{3,3} \end{bmatrix} \end{bmatrix}}_{\mathbf{R}}$$

$\mathbf{R}$ is an upper triangular matrix.

Based on the three equations above, $\mathbf{Q}$ is

$$\mathbf{Q} = [\mathbf{q}_1 \quad \mathbf{q}_2 \quad \mathbf{q}_3] = \begin{bmatrix} \sqrt{2}/2 & \sqrt{6}/6 & -\sqrt{3}/3 \\ \sqrt{2}/2 & -\sqrt{6}/6 & \sqrt{3}/3 \\ 0 & \sqrt{6}/3 & \sqrt{3}/3 \end{bmatrix}$$

We find that in this example, $\mathbf{Q}$ is an orthogonal matrix, because its column vectors are unit vectors and are pairwise orthogonal.

Based on the orthogonal projection relationships covered earlier in this section, the upper triangular matrix $\mathbf{R}$ can also be written as

$$\mathbf{R} = \begin{bmatrix} \mathbf{a}_1 \cdot \mathbf{q}_1 & \mathbf{a}_2 \cdot \mathbf{q}_1 & \mathbf{a}_3 \cdot \mathbf{q}_1 \\ 0 & \mathbf{a}_2 \cdot \mathbf{q}_2 & \mathbf{a}_3 \cdot \mathbf{q}_2 \\ 0 & 0 & \mathbf{a}_3 \cdot \mathbf{q}_3 \end{bmatrix} = \begin{bmatrix} \mathbf{q}_1^T @ \mathbf{a}_1 & \mathbf{q}_1^T @ \mathbf{a}_2 & \mathbf{q}_1^T @ \mathbf{a}_3 \\ 0 & \mathbf{q}_2^T @ \mathbf{a}_2 & \mathbf{q}_2^T @ \mathbf{a}_3 \\ 0 & 0 & \mathbf{q}_3^T @ \mathbf{a}_3 \end{bmatrix}$$

$\mathbf{R}$'s each column $\mathbf{r}_1$, $\mathbf{r}_2$, and $\mathbf{r}_3$ corresponds to $\mathbf{A}$'s $\mathbf{a}_1$, $\mathbf{a}_2$, and $\mathbf{a}_3$'s coordinates in $[\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3]$.

Finally, writing this as

$$A = QR$$

This is exactly the QR decomposition covered in the next section!



LA_09_04_02.ipynb uses a custom function to complete Gram-Schmidt orthogonalization of a matrix's column vectors; please study it on your own. In addition, please modify the code based on the equations above to recompute the matrix R .



Please use tools such as DeepSeek/ChatGPT to complete the following exercises for this section.

Q1. This section has only one exercise. We swap the values of the first two column vectors of the matrix above; the details are as follows. Please orthogonalize the following vectors based on the content covered earlier.

$$\mathbf{a}_1 = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}, \quad \mathbf{a}_2 = \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix}, \quad \mathbf{a}_3 = \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix}$$